

4th Semester		
ISCED Code	Course Code	Course Title
0612	CSE-2423	Database Management System
Credit Hours: 3	Contact Hours: 3	Type: Core, Engineering
Prerequisite: CSE-2424 (Database Management System Lab)		
Co-requisite: CSE-2424 (Database Management System Lab)		
Instructor's details, Consultation Hour and Office Locations Instructor: Dr. Mohammad Aman Ullah Associate Professor, Dept. of CSE, IIUC Contact: 01815641524 Office Location: Room # 405, Computer Science Academic Building Email: aman_cse@iiuc.ac.bd Consultation Hours: Saturday (10:00 – 12:30), Sunday (08:00 – 8:30), Tuesday (8:00 – 8:30)		

Course Assessments	CIE: Continuous Internal Evaluation	Attendance	10 Marks
		Class test/ Assignment/ Quizzes	10 Marks
		Mid-term	30 Marks
	SEE: Semester End Examination		50 Marks
	Total		100 Marks

Course Rationale:

The Database Management Systems (DBMS) course provides students with essential knowledge and skills to design, implement, and manage databases effectively. It explores fundamental concepts such as data modeling, relational database theory, SQL, and database architecture. This course equips learners to handle large volumes of data efficiently, ensuring data integrity, security, and accessibility. By understanding the role of databases in modern applications, students are prepared to meet the demands of various industries where data-driven decision-making and management are crucial.

Objectives:

Upon completion of the course, students will be able to:

1. Familiarize with Relational Databases, Database design, Data Storage and Querying, Transaction Management.
2. Ability to apply Relational Algebra, SQL, Query Optimization techniques, Data Integrity, Security, normalization techniques, Indexing Techniques, ACID Properties.
3. Design and Develop an enterprise data model that reflects the organization's goal.

Course Learning Outcomes (CLOs):

Upon successful completion of this course, students will be able to:

#	CLO Description	Weightage (%)
1.	Understand Relational Databases, Database design, Data Storage and Querying, Transaction Management.	40 %
2.	Apply Relational Algebra, SQL, Query Optimization techniques, Data Integrity, Security, normalization techniques, Indexing Techniques, ACID Properties.	20 %
3.	Create an enterprise data model that reflects the organization's fundamental business rules.	40 %

Mapping of CLO-PLO-DL-KP-EP-EA:

#	CLOs	PLOs	DL	KP	EP	EA	Teaching Learning Strategy (TLS)	Assessment Strategy (AS)
CLO1	Understand Relational Databases, Database design, Data Storage and Querying, Transaction Management.	PLO1	Cognitive/Understand	K2, K3	-	-	Lecture, Class discussion, Assignment, Note	Exam, Quiz, Assignment
CLO2	Apply Relational Algebra, SQL, Query Optimization techniques, Data Integrity, Security, normalization techniques, Indexing Techniques, ACID Properties.	PLO4	Cognitive/Analyze	K6	-	-	Lecture, Class discussion, Assignment, Note	Exam, Quiz, Assignment
CLO3	Create an enterprise data model that reflects the organization's fundamental business rules.	PLO3	Cognitive/Create	K5	-	-	Class discussion, Assignment	Exam, Project, Assignment
Note: DL: Domain/level of learning taxonomy, KP: Knowledge Profile, EP: Attribute of Complex Engineering Problems, EA: Attribute of Complex Engineering Activities, Learning Domains (C: Cognitive, A: Affective, P: Psychomotor)								

Course Content:

#	Content	Duration	CLOs
Mid-Term (30 Marks)			
1	Introduction: Database, data, database management system, Database system versus file system, Data model, Database language, Database user administration, Database system structure, Storage manager, Overview of Physical storage medium.	2 weeks	CLO1
2	Entity-Relationship Model: Entity sets, Relationship sets, Mapping Cardinalities, Keys, Attributes, Entity relationship diagram, Weak entity sets, Specialization, Generalization, Structure of Relational databases, Database Schema.	6 weeks	CLO3
3	The Relational Algebra and SQL: Selection, projection, Union, Set difference, Cartesian-product, Rename, Set-intersection, Natural-join, Division, Assignment, projection, Aggregate functions, Deletion, Insertion, Updating, Views, Nested sub-queries, Set membership, Set comparison.	4 weeks	CLO2
Final Exam: 50 Marks Group-A (20 Marks)			
4	Integrity, Security and Relational Database Design: Domain constraint, Integrity, Assertions, Triggers, Authorization, Authentication, Security, Privileges, Roles, and Audit trails, Encryption-Decryption Algorithm, Decomposition etc.	4 weeks	CLO2
5	Functional Dependency and Normalization: Functional Dependencies, Closure of a set of Functional dependencies. Un-normal Form (UNF), First Normal Form (1NF), Second Normal Form (2NF), Third Normal Form (3NF), Boyce and Codd Normal Form (BCNF).	4 weeks	CLO2
Group-B (30 Marks)			
6	Indexing and Hashing: Ordered indices, Hash indices, Hash function, Primary index, Secondary index, Dense, sparse, Multilevel indices, B+ tree index files, Handling Bucket Overflows, Overflow Chaining, Closed Hashing, Open Hashing, Linear probing, Hash indices, Dynamic Hashing.	4 weeks	CLO2
7	Transaction:	4 weeks	CLO2

	ACID Properties, Transaction state diagram, Implementation of Atomicity and Durability, Shadow copy technique, Concurrent Execution, Serializability, Recoverability, Recoverable schedule, Cascadeless Schedules, Implementation in Isolation, Testing of Serializability.		
8	Concurrency control, Recovery System and Distribute databases: Lock-Based Protocols, Granting of locks, Two-phase locking protocol, Graph based protocol, Tree protocol, Timestamp based protocols, Deadlock detection and recovery. Failure classification, Storage types, Checkpoints. Distributed data, Replication and Fragmentation.	2 weeks	CLO2

Weekly Activity Plan:

Week	Topic	TLS	AS	CLOs
Week 1	Introduction: Database, data, database management system, Database system versus file system, Data model, Database language, Database user administration	Lecture		CLO1
Week 2	Introduction Database system structure, Storage manager, Overview of Physical storage medium. Entity-Relationship Model: Entity sets, Relationship sets, Mapping Cardinalities, Keys, Attributes, Entity relationship diagram, Weak entity sets.	Lecture, Class discussion		CLO1, CLO3
Week 3	Entity-Relationship Model: Specialization, Generalization, Structure of Relational databases, Database Schema.	Lecture	Quiz	CLO3
Week 4	The Relational Algebra and SQL: Selection, projection, Union, Set difference, Cartesian-product, Rename, Set-intersection, Natural-join, Division	Lecture, Class discussion		CLO2
Week 5	The Relational Algebra and SQL: Assignment, projection, Aggregate functions, Deletion, Insertion, Updating, Views, Nested sub-queries, Set membership, Set comparison.	Lecture	Exam	CLO2
Week 6	Review of the midterm exam content			
Week 7	Integrity, Security and Relational Database Design:	Lecture		CLO2

Week	Topic	TLS	AS	CLOs
	Domain constraint, Integrity, Assertions, Triggers, Authorization, Authentication			
Week 8	Integrity, Security and Relational Database Design: Security, Privileges, Roles, and Audit trails, Encryption-Decryption Algorithm, Decomposition etc. Functional Dependency and Normalization: Functional Dependencies, Closure of a set of Functional dependencies.	Lecture, Class discussion		CLO2
Week 9	Functional Dependency and Normalization: Un-normal Form (UNF), First Normal Form (1NF), Second Normal Form (2NF), Third Normal Form (3NF), Boyce and Codd Normal Form (BCNF).	Lecture	Quiz	CLO2
Week 10	Indexing and Hashing: Ordered indices, Hash indices, Hash function, Primary index, Secondary index, Dense, sparse, Multilevel indices, B+ tree index files	Lecture, Class discussion, notes		CLO2
Week 11	Indexing and Hashing: Handling Bucket Overflows, Overflow Chaining, Closed Hashing, Open Hashing, Linear probing, Hash indices, Dynamic Hashing. Transaction: ACID Properties, Transaction state diagram, Implementation of Atomicity and Durability, Shadow copy technique	Lecture, Class discussion		CLO2
Week 12	Transaction: Concurrent Execution, Serializability, Recoverability, Recoverable schedule, Cascadeless Schedules, Implementation in Isolation, Testing of Serializability. Concurrency control, Recovery System and Distribute databases: Lock-Based Protocols, Granting of locks, Two-phase locking protocol, Graph based protocol, Tree protocol, Timestamp based	Lecture, Class discussion	Assignment	CLO2

Week	Topic	TLS	AS	CLOs
	protocols			
Week 13	Concurrency control, Recovery System and Distribute databases: Deadlock detection and recovery. Failure classification, Storage types, Checkpoints. Distributed data, Replication and Fragmentation.	Lecture, Class discussion		CLO2
Week 14	Concurrency control, Recovery System and Distribute databases: Distributed data, Replication and Fragmentation.	Lecture	Exam	CLO2
Week 15	Review of the semester final content			

Marks Distribution:

Course Assessment Pattern (Theory courses):

		CIE (50 marks)			SEE (50 marks)
Cognitive learning	Affective learning	Attendance Marks (10)	Assignment/ Class Test (10)	Mid-term (30)	Written Exam (50)
Remember					5
Understand			5	5	15
Apply					
Analyze			5	15	30
Evaluation					
Create				10	
	Responding	10			
Total allocated marks		10	10	30	50

Grading Policy: As per IIUC grading policy

Learning Materials:

Text Books:

1. Abraham Silberschatz, Henry F. Korth, S. Sudarshan, "Database System Concept", 7th Edition, McGraw-Hill, 2013, ISBN : 978-0-07-352332-3.
2. RamezElmasri, Shamkant B. Navathe, Fundamentals of database systems, 6th Pearson Education, 2011, ISBN : 10: 987-0-136-08620-9.

Reference Books:

1. Hector Garcia-Molina, Jeff Ullman, and Jennifer Widom Database Systems: The Complete Book, 3rd Edition, Pearson (Addison-Wesley + Prentice-Hall) 2008, 9870138613370.
2. Connolly, Thomas M., Begg, Carolyn E. Database systems : a practical approach to design, implementation, and management, 4th, Addison-Wesley, 2005, 987321210255.